From Replica to Reference: Active Learning in Engineering Education for Digital Twin–Guided Real-System Evolution

João Mello da Silva[1], Mateus Halbe Torres[1], Edison Ishikawa[1], Ari Melo Mariano[1], César Antônio Ciuffo Moreira[2]

[1] University of Brasília-UnB, Brazil
[2] APEX Brasil, Brazil

Email: joao.mellos@gmail.com, mateusconsigo@gmail.com, edisonishikawa@gmail.com, arimariano@unb.br, cesar.moreira@apexbrasil.com.br

Abstract

This paper reports an active-learning application in a graduate IT project management course through a paradigm shift: treating the Digital Twin (DT) as the reference model that drives real-system (Real-life Twin=RT) evolution, rather than a subordinate replica validated against the present. Having PMBOK Guides and PMI BA Guide as major project management references, the course focuses on developing a project management guide tailored for DT components of Brasilia's Solid Waste System, in an adapted PMBOK extension format. Distributed in groups, students work with a paired representation of the Real System in Question (RSQ=RT) and its digital counterpart (DSQ=DT), initially describing both systems in terms of Function, Structure, Process, and Context (F/S/P/C), showing interactions among these dimensions. To structure what the DT provides, learners apply the Digital Twin Consortium Capabilities (DTC) Periodic Table (WHAT: DS–Data Services, IR–Integration, IC–Intelligence, UX–User Experience, and MG–Management and TW–Trustworthiness). After structuring how capabilities are put together, the groups map selected functions to DTC AI-agent capabilities (HOW: PW-Perception/Knowledge, CG-Cognition/Reasoning, LA-Learning/Adaptation, IC-Interaction/Collaboration, AE-Action/Execution, and GS-Governance/Safety), spanning conversational, workflow, autonomous, and collaborative agent modalities. Learning activities are organized as twin-in-the-loop decision cycles: teams use DSQ to generate and compare implementable configuration chains DSQ(t_1), DSQ(t_2), ..., DSQ(t_n) that target a desired future state RSQ(t_n), with each update explicitly informing the required portfolio/program/project/extension/ operations effort in the real system. Evaluation triangulates rubric-based assessment of artifacts (traceability from F/S/P/C to WHAT/HOW choices, uncertainty and risk treatment, and decision defensibility), concept measures of anticipatory systems reasoning, and reflective analyses of reference-direction change (real to twin vs. twin to real). Expected results include a transferable instructional pattern for AI-enabled, governance-aware digital twins and empirical insight into how "twin-first" framing strengthens risk-informed, future-oriented engineering judgment.

Keywords: Active Learning; Digital twins; Anticipatory systems; Exploring New .

1 Introduction

Digital twins (DTs) have matured from "models that resemble reality" into socio-technical systems that coordinate data, integrations, analytics, interfaces, governance, and trust to support decisions and execution. The Digital Twin Consortium (DTC) emphasizes DT as a virtual representation of entities and processes synchronized with its real counterpart at specified frequency and fidelity (Digital Twin Consortium, 2025). In educational practice, however, DTs are still frequently framed as subordinate replicas, where the primary success criterion is to mirror the present state. That framing is limiting because it under-weights the central reason DTs matter in engineering and operations: they can be used to steer systems toward desired futures under constraints, uncertainty, and risk.

This paper reports the initial phase, corresponding to the description of both, the Real System in Question (Real-life Twin=RSQ=RT) and its corresponding Digital System in Question (Digital Twin=DSQ=DT), of an

active-learning design for a graduate IT project management course that makes a deliberate reference-direction change. Having PMBOK4, PMBOK6, PMBOK7 and PMBOK8 as major project management references, the course focus is the development of a project management guide tailored for DT components of Brasilia's Solid Waste System, in an adapted PMBOK extension format. Instead of "Real-life Twin to Digital Twin" validation being the main loop, students are guided to practice "Digital Twin to Real-life Twin" reasoning: the DT becomes the reference model that drives real-system evolution over time. The course context reinforces that the output cannot be merely conceptual; it must be execution-aware and governance-aware, explicitly connecting Digital System in Question (DSQ) decisions to portfolio, program, project/extensions, and operations work (Project Management Institute, 2008, 2017a, 2017b, 2021, 2025; University of Brasília, 2025a). In this approach, students are assessed not by how sophisticated their models look, but by how defensible their decisions are under uncertainty, and how implementable their evolution pathway is.

2 Background and Conceptual Foundations

2.1 From digital replica to decision reference

A DT becomes strategically meaningful when it helps decision-makers choose among alternatives and justify change sequencing. When students treat DT as a replica, their work tends to optimize fidelity against today's conditions; when they treat DT as a reference, their work tends to optimize decision defensibility toward a future target. This matters because real systems are not static: constraints change, operations evolve, risks materialize, and the information environment shifts. Treating the DT as a reference forces students to articulate a desired future state and make the pathway explicit, rather than implicitly assuming that improvement is the natural consequence of measurement.

2.2 RSQ/DSQ and F/S/P/C traceability

To keep the pattern auditable, students build a paired representation concept of Real Systems in Question (RSQ) and DSQs. Both are described using Function, Structure, Process, and Context (F/S/P/C), making explicit how changes in context propagate into feasible structures, processes, and functions (Gharajedaghi, 2011). The course materials provide the conceptual scaffolding and deliverable expectations for this representation, emphasizing traceability and disciplined articulation of system boundaries and requirements (University of Brasília, 2025a, 2025b). F/S/P/C plays the role of the "argument spine" of the work: capabilities, agent functions, and implementation steps must remain consistent with this system description.

2.3 Capability logic: WHAT and HOW

The pattern uses a two-layer capability logic to avoid tool-first thinking. First, students decide WHAT the twin must provide using the Digital Twin Consortium Capabilities Periodic Table categories (Figure 1), which encompass DS (Data Services), IR (Integration), IC (Intelligence), UX (User Experience), MG (Management), and TW (Trustworthiness) (Digital Twin Consortium, 2025). Consistent with the table's framing, these categories can be read as a value progression from DATA to INFORMATION, KNOWLEDGE, KNOW-HOW and, ultimately, BUSINESS WISDOM.

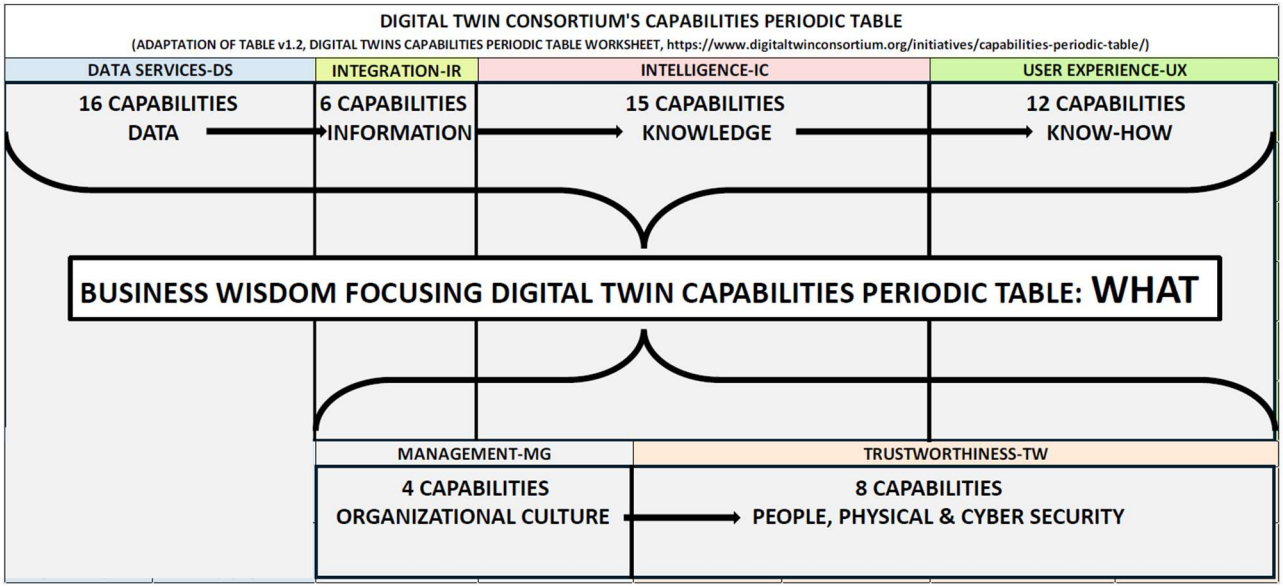


Figure 1. Digital Twin Capabilities Periodic Table, WHAT: Data/Information/Knowledge/Know-how/Wisdom

Second, students decide HOW selected functions will be performed using the AI Agent Capabilities Periodic Table categories (Figure 2) in the same format: PW (Perception & Knowledge), CG (Cognition & Reasoning), LA (Learning & Adaptation), AE (Action & Execution), IC (Interaction & Collaboration), and GS (Governance & Safety) (Digital Twin Consortium, 2025). This separation is especially important in educational settings where generative AI can bias teams toward solutions that sound powerful but have unclear requirements, weak governance, or poor execution realism. It is worthwhile emphasizing the role of the Organizational Performance Matrix Readiness x Excellence of showing the GS-Governance & Safety capabilities organizational excellence against the IC-Interaction & Collaboration capabilities organizational readiness.

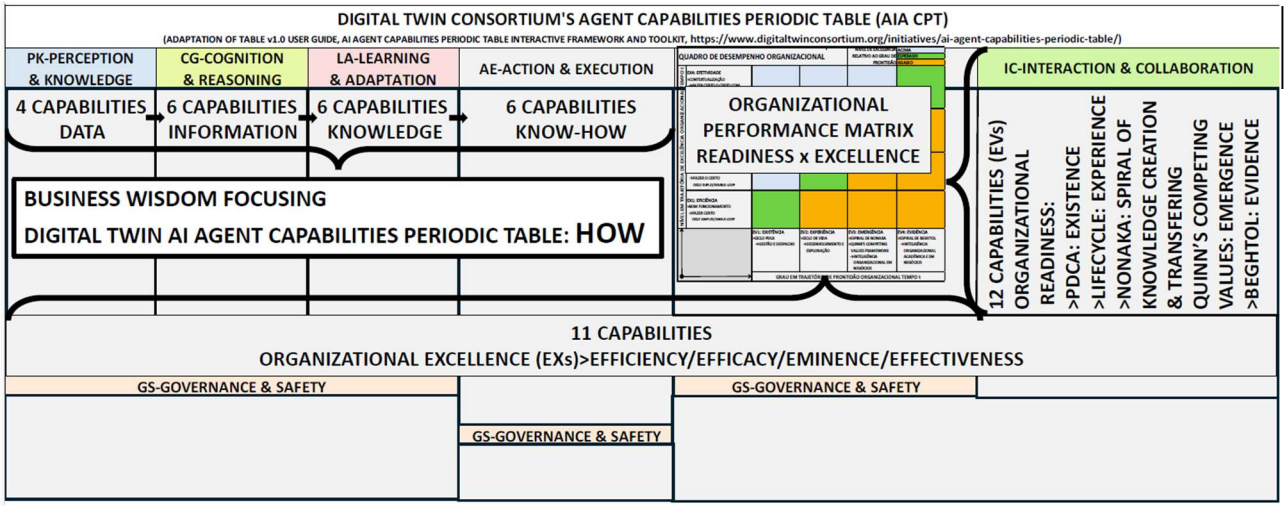


Figure 2. Digital Twin Capabilities Periodic Table, HOW: Data/Information/Knowledge/Know-how/Wisdom

Figure 3 summarizes the two-layer capability logic used in the instructional pattern. At the WHAT (Twin) layer, teams make a requirements-oriented selection of digital twin capability categories (DS/IR/IC/UX/MG/TW) to justify what the DSQ must provide to support decisions (Digital Twin Consortium, 2025). At the HOW (Twin/Agent) layer, teams make a requirements-oriented selection of AI-agent capability categories (PW/CG/LA/AE/IC/GS) to justify how selected DSQ functions will be implemented to support decisions, with the choice of agent modality (conversational, workflow, autonomous, or collaborative) explicitly constrained by governance and safety requirements (Digital Twin Consortium, 2025).

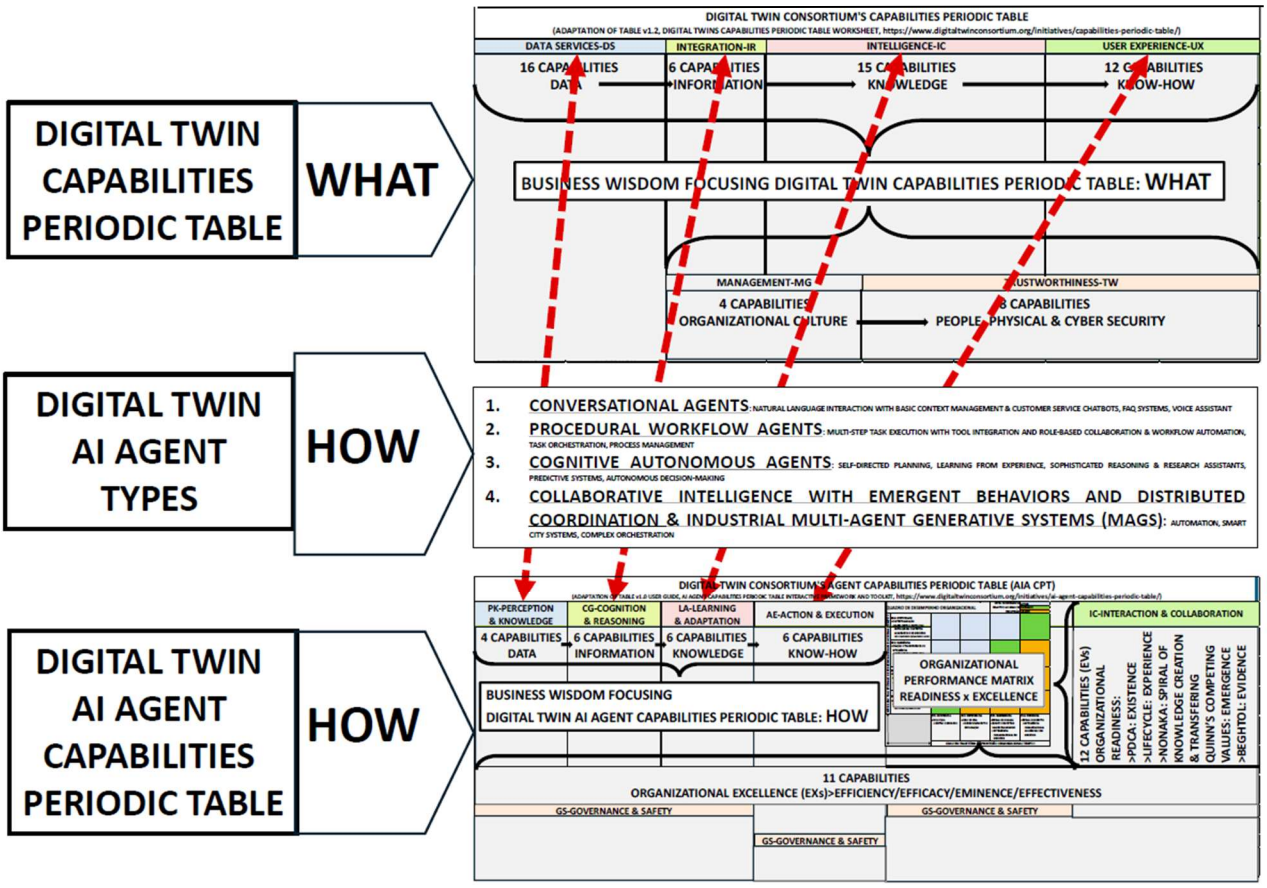


Figure 3. Digital Twin Capabilities Periodic Table: WHAT (Capabilities), HOW (AI Types), HOW (Capabilities)

2.4 IT/OT/ET convergence and why it is central to "Digital Twin-first"

DTs become most challenging and most valuable precisely when they sit at the intersection of IT (Information Technology systems and data/analytics), OT (Operational Technology such as sensors, automation, and industrial control), and ET (Engineering Technology such as models, design intent, and system engineering artifacts) (Hollywood, 2020). In practice, the DT must reconcile heterogeneous data sources and time scales: IT often operates on business rhythms and transactional correctness; OT operates on real-time constraints, safety, and reliability; ET operates on design fidelity, physics, and lifecycle reasoning. When these worlds converge, requirements also converge: integration and data services cannot be separated from trustworthiness, and intelligence cannot be separated from governance and operational safety (Digital Twin Consortium, 2025; Hollywood, 2020).

The instructional design leverages this convergence to make students practice realistic reasoning. RSQ/DSQ boundaries must explicitly include where IT ends and OT begins (e.g., what data is available, at what cadence, under what security constraints) and how ET representations are maintained (e.g., whether the DSQ includes engineering models, assumptions, and their versioning). As shown in Figure 4, the "Digital Twin-first" framing then becomes a disciplined way to coordinate convergence decisions: teams use DSQ(t1), DSQ(t2), ..., DSQ(tn) to make explicit which integration and trust steps are prerequisites for higher-level intelligence and agent autonomy, and which parts of the system must remain human-governed due to safety or ethical constraints.

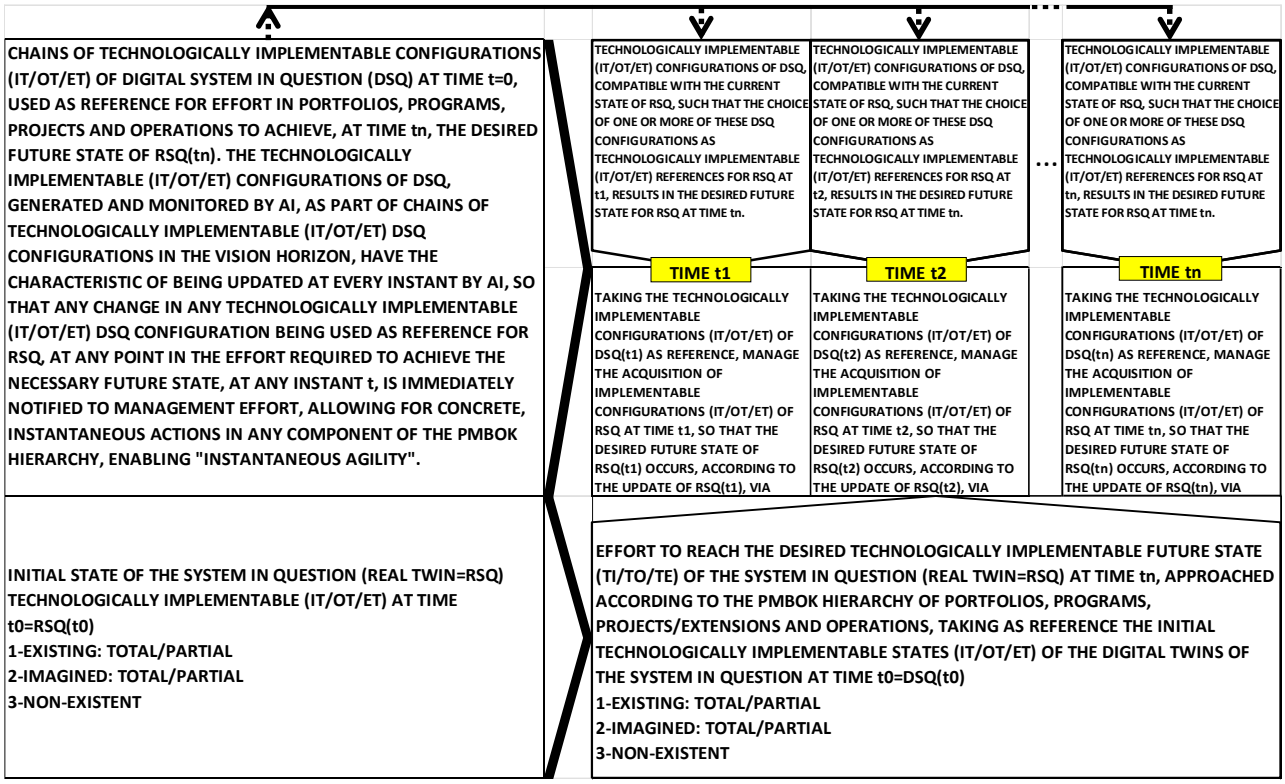


Figure 4. Digital Twin as Reference for Real Twin

3 Instructional Design

3.1 Course environment and active-learning structure

The pattern is designed for a graduate IT project management course in which teamwork is the default learning unit and active learning is operationalized through structured workshops, problem-based learning, and student-led artifact production (Guerra et al., 2017; University of Brasília, 2025a). Students are not treated as passive recipients of frameworks; instead, they use frameworks to build a coherent, auditable argument about system evolution. The pattern is compatible with the course's broader emphasis on disciplined execution and managerial reasoning, using PMI project management guidance as a shared language for packaging work and defending decisions (Project Management Institute, 2008, 2017a, 2017b, 2021, 2025).

3.2 Twin-in-the-loop decision cycles and configuration chains

The learning workflow is organized as iterated twin-in-the-loop decision cycles. Teams first are exposed to an established RSQ(t0) baseline and define DSQ(t0) as the initial twin scope with explicit synchronization cadence and fidelity. They then define a desired future state RSQ(tn) and work backward through implementable steps. Each step proposes a DSQ configuration DSQ(tk) that enables specific decision functions and governance constraints and then derives the corresponding portfolio/program/project/extension/operations work package required to produce the real-system changes toward RSQ(tk). Over multiple iterations, the chain DSQ(t1), DSQ(t2), ..., DSQ(tn) becomes the structured AI artifact that encodes sequencing, dependencies, trade-offs, and governance gates.

This structure is intentionally "execution coupled." Instead of allowing students to design an idealized twin, each DSQ step must explain what will change in real operations, which stakeholders are involved, what risks are introduced or mitigated, and what trustworthiness and governance measures are required. This ties

directly to the project's management intent and aligns with the course's deliverable framing (University of Brasília, 2025b).

3.3 A transferable worked-example template

To support transfer across domains, teams instantiate a worked-example template. For the course being reported in this paper, the concrete RSQs are the components of the Solid Waste System in Brasilia, Brazil. Specifically for this case, the components of the RSQ were categorized in terms of four functional elements: Selective Collection, Triage, Transformation and Commercialization. For any context, the template always begins with a clear RSQ scope definition (including stakeholders and constraints), then defines DSQ scope and fidelity, selects WHAT capabilities are required for decisions, maps HOW capabilities to agent functions, and finally produces a sequenced configuration chain with packaging and risk treatment. While the domain can vary widely (e.g., urban infrastructure, industrial process improvement, service systems), the learning objective is invariant: disciplined, future-oriented, governance-aware decision-making through a DT-as-reference lens.

As an input to the framework shown in Figure 4, the course flow begins with a review of PMBOK8, followed by the evolution of PMBOK8 into an adapted PMBOK extension format (PMBOK8 to GPH.SIPA.S), through the separate inclusion, via teamwork in "partial colleges" dedicated to each SIPA dimension (S-Sustainability, I-Innovation, P-Privacy, and A-Business Analysis), so that tailoring becomes an explicit methodological step rather than an informal adaptation. This PMBOK8 evolution, in an adapted PMBOK extension format is then advanced, through teamwork in a "total college", moving from partial outputs obtained by the "partial colleges" by dimension SIPA, to a consolidated version involving all SIPA dimensions and bridging toward PMBOK8 framing (GPH.SIPA.S+), while remaining anchored in operational stakeholder perspectives (e.g., product developers, waste-picker cooperatives, and landfill operators). In parallel, the RSQ is framed as both a production system and a human activity system by locating solid waste within the broader sanitation system and structuring the solid-waste domain through P3RD priorities (P-Prevention, R1-Reduction, R2-Reuse, R3-Recycling, DS-Disposal), complemented by the "Digital Mobilizer" role that connects operational reality to digital project framing.

4 Learning Artifacts and Deliverables

The deliverables in this pattern are designed to function as a single integrated argument rather than separate assignments. The RSQ/DSQ F/S/P/C descriptions provide the system-level grounding and boundary clarity; the WHAT selection provides a requirements rationale; the HOW mapping provides an implementation rationale; and the DSQ configuration chain provides the evolution narrative and sequencing logic. Students are expected to ensure that artifacts remain mutually consistent: capabilities must be traced to system needs, integration and governance must be explicitly addressed, and the path from DSQ decisions to real-world execution must be defensible.

Because the twin is treated as a decision reference, the quality of deliverables is judged primarily by traceability and defensibility rather than by presentation polish. A strong submission makes explicit how context constraints (e.g., privacy, safety, procurement) shape feasible structures and processes; it uses the WHAT layer to explain why a capability is needed; it uses the HOW layer to justify which agent modalities are appropriate; and it translates the resulting DSQ(t1), DSQ(t2), ..., DSQ(tn) chain into implementation work that can be managed under realistic organizational constraints.

5 Assessment and Evaluation

Evaluation triangulates three evidence streams: rubric-based assessment of artifacts, concept measures of anticipatory reasoning, and reflective analyses of the reference-direction shift. The artifact rubric evaluates traceability from F/S/P/C to WHAT/HOW choices, coherence of system boundaries, explicit uncertainty and risk treatment, and decision defensibility. To operationalize this, the rubric uses a 0–4 scale per dimension (0 absent, 1 weak, 2 adequate, 3 strong, 4 exemplary) across the following dimensions: (i) RSQ/DSQ boundary clarity and F/S/P/C completeness; (ii) traceability from F/S/P/C to WHAT selections; (iii) coherence between WHAT selections and HOW mapping to agent capabilities; (iv) quality of DSQ configuration chain sequencing and dependency reasoning; (v) explicit risk, uncertainty, and governance treatment; and (vi) implementability of the portfolio/program/project/extension/operations packaging (Project Management Institute, 2008, 2017, 2021, 2025; University of Brasília, 2025a, 2025b). Instructors and/or teaching assistants can improve scoring consistency by calibrating on two anchor submissions and discussing borderline cases before grading.

Concept measures are used to check whether students can reason about futures rather than merely describe present conditions. Typical prompts include identifying leading indicators and feedback loops, articulating decision criteria under uncertainty, and distinguishing between structural constraints and process variability. Reflection is assessed through a structured memo in which students compare real to twin validation logic against twin to real guidance logic and discuss how the change affects accountability, governance, and risk posture. Risk management reasoning is supported by established risk frameworks and project management guidance (International Organization for Standardization, 2018, 2019; Project Management Institute, 2017, 2021; University of Brasília, 2025b).

6 Discussion

The reference-direction shift changes what students optimize for. In a replica-first mindset, the dominant criteria become representational fidelity of the present; in a reference-first mindset, the dominant criteria become defensibility of a future-oriented pathway under change. This naturally elevates integration and trustworthiness from optional concerns to gating concerns: in converged IT/OT/ET systems, the ability to reason well depends on whether the DSQ has access to trusted data, robust integrations, and governance mechanisms that prevent unsafe or unethical actions. The WHAT/HOW separation reinforces this, because teams must make requirements explicit and then justify which agent modalities can be responsibly deployed given the system's constraints and risk posture.

The DSQ configuration chain is also pedagogically important because it externalizes sequencing, dependencies, and decision trade-offs. Students learn that autonomy is not a switch; it is an incremental capability that depends on integration maturity, data quality, operational constraints, and governance readiness. This is precisely where IT/OT/ET convergence becomes an educational advantage: it forces teams to confront that operational systems are not merely data problems and that engineering intent is not merely documentation. The twin-in-the-loop cycles then become a method for reasoning about the interplay between operational risk, technical feasibility, and managerial execution.

Limitations follow from the nature of design reports and course-based studies. Outcomes depend on cohort maturity, domain choice, and the extent to which teams have access to realistic data and operational stakeholders. If empirical learning gains are claimed, future work should include multi-cohort replication, inter-rater reliability checks for rubric scoring, and validation of the anticipatory reasoning instrument. A promising extension is to analyze which capability categories (especially TW in WHAT and governance/safety in HOW) most strongly correlate with improvements in decision defensibility and sequencing realism.

7 Conclusion

This paper reported the initial phase, corresponding to the description of both, the Real System in Question

(Real-life Twin=RSQ=RT) and its corresponding Digital System in Question (Digital Twin=DSQ=DT), of an active-learning instructional design that deliberately moves the digital twin from subordinate replica to decision reference, treating it as a driver of real-system evolution rather than merely an after-the-fact mirror. The design combines RSQ/DSQ paired representation using F/S/P/C, a two-layer capability logic (WHAT via the DTC Capabilities Periodic Table and HOW via AI-agent capability families), and twin-in-the-loop decision cycles that generate an implementable DSQ configuration chain targeting a desired future RSQ state. By explicitly foregrounding IT/OT/ET convergence, the pattern forces students to address integration, trustworthiness, governance, and sequencing realism as central design constraints, strengthening risk-informed and future-oriented engineering judgment in an execution-coupled educational setting.

Therefore, repeating the left hand-side of Figure 4: Chains of technologically implementable configurations (IT/OT/ET) of Digital System in Question (DSQ) at time t=0, used as reference for effort in portfolios, programs, projects and operations to achieve, at time tn, the desired future state of RSQ(tn). The technologically implementable (IT/OT/ET) configurations of DSQ, generated and monitored by AI, as part of chains of technologically implementable (IT/OT/ET) DSQ configurations in the vision horizon, have the characteristic of being updated at every instant by AI, so that any change in any technologically implementable (IT/OT/ET) DSQ configuration being used as reference for RSQ, at any point in the effort required to achieve the necessary future state, at any instant t, is immediately notified to management effort, allowing for concrete, instantaneous actions in any component of the PMBOK hierarchy, enabling "instantaneous agility".

Based on the digital twin approach described in this paper, the authors intend to conduct surveys with the students to assess actual improvement and evolution of learners' abilities regarding engineering decision-making and thinking.

8 Acknowledgements

This work was partially developed in the context of project 2023-1-DK01-KA220-HED-00165709, "EGALITARIAN - Education, Digitalization and Collaboration for Sustainability" which has been funded with support from the European Commission-EC. This publication reflects the views only of the authors, and the EC cannot be held responsible for any use which may be made of the information contained therein.

References

Digital Twin Consortium. (2025). AI agent capabilities periodic table – interactive framework and toolkit. Retrieved February 8, 2026, from https://www.digitaltwinconsortium.org/initiatives/ai-agent-capabilities-periodic-table/

Digital Twin Consortium. (2025). Data and knowledge management in digital twin systems. Retrieved February 8, 2026, from https://www.digitaltwinconsortium.org/

Digital Twin Consortium. (2025). Digital twin consortium capabilities periodic table – worksheet (v1.2).

Gharajedaghi, J. (2011). Systems thinking: Managing chaos and complexity: A platform for designing business architecture (3rd ed.). Morgan Kaufmann.

Guerra, A., et al. (2017). Latin case [Online PDF]. Retrieved February 8, 2026, from https://vbn.aau.dk/ws/portalfiles/portal/262849868/Latin_Case_online.pdf

Hollywood, P. (2020). Why convergence of IT/OT needs to include ET. ARC Advisory Group. Retrieved February 9, 2026, from https://www.arcweb.com/industry-best-practices/why-convergence-itot-needs-include-et

International Organization for Standardization. (2018). ISO 31000 – Risk management.

International Organization for Standardization. (2019). ISO 31010 – Risk management – Risk assessment techniques.

Project Management Institute. (2008). A guide to the project management body of knowledge (PMBOK guide) (4th ed.).

Project Management Institute. (2017a). A guide to the project management body of knowledge (PMBOK guide) (6th ed.).

Project Management Institute. (2017b). The PMI Guide to Business Analysis.

Project Management Institute. (2021). A guide to the project management body of knowledge (PMBOK guide) (7th ed.).

Project Management Institute. (2025). A guide to the project management body of knowledge (PMBOK guide) (8th ed.).

University of Brasília. (2025a). Gpti course plan (2025s) – graduate it project management (ppca/unb) [Internal course material].

University of Brasília. (2025b). Gpti task summary (2025s) – deliverables, capability tables, and evaluation [Internal course material].